THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Final Exam, 2022

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Reading time: 30 minutes — Writing time: 3 hours — Upload time: 30 minutes

This exam consists of 19 pages (including this page) with 11 questions and 106 total marks

Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier and blank loose-leaf paper.
- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted. No headphones or earphones are permitted.

Instructions to Students

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- You must not be out of webcam view at any time without supervisor permission.
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- If you are writing answers on the exam or masked exam and need more space, use blank paper. Note this in the answer box, so the marker knows.
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Question 1 (21 marks)

Short questions: In addition to your answer, you must show your working.

(a) Solve the following initial value problem

$$F(x) = \int \left(\sqrt[3]{x} + \cos(\pi x)\right)\, dx \quad \text{with } F(0) = 1$$

$$F(x) = \int \left(x^{\frac{1}{3}} + \cos(\pi x)\right)\, dx = \frac{3}{4}x^{\frac{4}{3}} + \frac{\sin(\pi x)}{\pi} + C \quad \boxed{1}$$

$$F(0) = \frac{3}{4}0^{\frac{4}{3}} + \frac{\sin(0)}{\pi} + C \quad \boxed{1\,\mathsf{M}}$$

$$1 = 0 + C$$

$$C = 1$$

Hence, $F(x) = \frac{3}{4}x^{\frac{4}{3}} + \frac{\sin(\pi x)}{\pi} + 1$. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

(b) You have 5 different pictures. How many different arrangements can be made by hanging them in a row, if at least three pictures are used for each arrangement?

We know we can have arrangements of 3, 4 or 5 pictures in a row. Since these options are mutually exclusive, addition principle applies and so:

$$\#\text{ of arrangements} = \#(\text{arr. of } 3) + \#(\text{arr. of } 4) + \#(\text{arr. of } 5) \quad \boxed{1\,\mathsf{M}}$$

Each of the options above consists of permutations of 3, 4 or 5 elements from a set of 5, so we have:

$$\begin{aligned}\#\text{ of arrangements} &= \#(\text{arr. of } 3) + \#(\text{arr. of } 4) + \#(\text{arr. of } 5)\\ &= (5\times 4\times 3) + (5\times 4\times 3\times 2) + (5\times 4\times 3\times 2\times 1) \quad \boxed{1\,\mathsf{M}}\\ &= 60 + 120 + 120\\ &= 300\end{aligned}$$

A total of 300 arrangements can be made $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

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(c) Find the equation of the tangent line to $f(x) = e^{x^2+x}$ about $x = 0$.

Work out each component separately

$$f(x) = e^{x^2+x} \qquad f(0) = e^0 = 1 \ \boxed{1}$$
$$f'(x) = (2x+1)e^{x^2+x} \qquad f'(0) = (0+1)e^0 = 1 \ \boxed{1}$$

So the tangent line to $f(x)$ about $x = 0$ is:

$$\begin{aligned} y - f(0) &= f'(0)(x-0) \\ y &= f(0) + f'(0)(x-0) \\ y &= 1 - 1(x-0) \\ y &= 1 - x \ \boxed{1\,\mathsf{A}} \end{aligned}$$

Subtotal 3

(d) Let $f(x) = 3x^2 \log(6x)$. Find $f'(x)$.

Use product rule $f = u \cdot v$: $\boxed{1\,\mathsf{M}}$(product rule)

$$u = 3x^2 \qquad u' = 6x \quad \boxed{1}\text{(correct } u' \text{ and } v'\text{)}$$
$$v = \log(6x) \qquad v' = \frac{1}{6x} \times 6 = \frac{1}{x} \text{ (by chain rule)}$$

So $f'(x) = 6x\log(6x) + 3x^2\frac{1}{x} = 6x\log(6x) + 3x = 3x(2\log(6x)+1)$ $\boxed{1\,\mathsf{A}}$

Subtotal 3

(e) Evaluate the following:

$$\int \frac{x^5 + 2x^3 + x}{x^4}\, dx$$

$$\int \frac{x^5 + 2x^3 + x}{x^4}\, dx = \int \left(x + 2\frac{1}{x} + x^{-3}\right) dx \ \boxed{1\,\mathsf{M}}$$

Therefore, $\int \frac{x^5+2x^3+x}{x^4}\, dx = \frac{x^2}{2} + 2\log|x| - \frac{1}{2x^2} + C$

$\boxed{2\,\mathsf{A}}$ (correct integral + adding C)

Subtotal 3

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(f) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x\to 2} \frac{x^3 - 7x + 6}{x - 2}$$

At $x = 2$, the numerator is $(2)^3 - 7(2) + 6 = 0$ and the denominator is $2 - 2 = 0$. So we have to simplify

$$\begin{array}{r|l} & x^2 + 2x - 3 \quad \boxed{1\,\mathsf{M}} \\ \hline x - 2 & x^3 \qquad\quad - 7x + 6 \\ & -x^3 + 2x^2 \\ \hline & \quad 2x^2 - 7x \\ & \quad -2x^2 + 4x \\ \hline & \qquad -3x + 6 \\ & \qquad 3x - 6 \\ \hline & \qquad\quad 0 \end{array}$$

So

$$\begin{aligned} \lim_{x\to 2} \frac{x^3 - 7x + 6}{x - 2} &= \lim_{x\to 2} \frac{(x-2)(x^2 + 2x - 3)}{x - 2} \\ &= \lim_{x\to 2} (x^2 + 2x - 3) \quad \boxed{1} \\ &= 2^2 + 2 \cdot 2 - 3 = 5. \quad \boxed{1\,\mathsf{A}} \end{aligned}$$

Subtotal 3

(g) Consider the quadratic function $f(x) = x^2 + 5x + 2$. Write $f(x)$ in turning-point form (i.e. $f(x) = (x - h)^2 + k$) and hence determine the coordinates of its turning point.

We complete the square: $\boxed{1\,\mathsf{M}}$

$$\begin{aligned} f(x) &= x^2 + 5x + 2 \\ &= x^2 + 2\left(\frac{5}{2}\right)x + \left(\frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + 2 \\ &= \left(x + \frac{5}{2}\right)^2 - \frac{25}{4} + 2 \\ &= \left(x + \frac{5}{2}\right)^2 - \frac{17}{4} \quad \boxed{1\,\mathsf{A}} \end{aligned}$$

So $f(x)$ has a turning point at $\left(-\frac{5}{2}, -\frac{17}{4}\right)$ $\boxed{1\,\mathsf{A}}$

Subtotal 3

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Extended problems: Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justification of your solutions.

Question 2 (13 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$, with $f(x) = 2\sin(2x) + 1$.

(a) State the domain and range of the function. You should justify your answer for the range - you may do so by indicating what graph transformations you need to transform $\sin(x)$ into the given function $f(x)$.

The domain of the function is $D = \mathbb{R}$. 1 A
The function $f(x)$ can be obtained from $\sin(x)$, which has range $[-1, 1]$, via: 1 J
(i) a compression by a factor of 2 in the x-direction (which doesn't affect the range),
(ii) a streching by a factor of 2 in the y-direction (which changes the range to $[-2, 2]$) and
(iii) a vertical shift upwards by 1 unit (which changes the range to $[-1, 3]$).
Therefore, the range of the function is $R = [-1, 3]$ 1 A

Subtotal 3

(b) Is $f(x)$ injective? Justify your answer.

The function is not injective. 1 A
For example, take $x = 0$ and $y = \pi$. Clearly $x \neq y$ but
$f(x) = f(0) = 2\sin(0) + 1 = 1 = 2\sin(2\pi) + 1 = f(\pi) = f(y)$,
so $f(x)$ is not injective.
1 J (a valid counterexample or explanation)

Subtotal 2

(c) Find the y-intercept of $f(x)$.

To find the y-intercept, we substitute $x = 0$: 1 M
$f(0) = 2\sin(0) + 1 = 1$,
so the y-intercept is $(0, 1)$. 1 A

Subtotal 2

(d) Find all the x-intercepts of the function $f(x)$.

To find all the x-intercepts, we need to find all solutions of the equation $f(x) = 0$ (i.e. we need to solve $2\sin(2x) + 1 = 0$) $\boxed{1\,\mathsf{M}}$

$$\begin{aligned} 2\sin(2x) + 1 &= 0 \\ 2\sin(2x) &= -1 \\ \sin(2x) &= -\frac{1}{2} \end{aligned}$$

This implies that the angle in the range $[0, 2\pi)$ is $\frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\boxed{2}$
Therefore, the general solution is:

$$2x = \frac{\pi}{6} + 2k\pi, \text{ or } 2x = \frac{5\pi}{6} + 2k\pi, \qquad k \in \mathbb{Z} \ \boxed{1\,\mathsf{M}}$$

$$x = \frac{\pi}{12} + k\pi, \text{ or } x = \frac{5\pi}{12} + k\pi, \qquad k \in \mathbb{Z} \ \boxed{1\,\mathsf{A}}$$

so $f(x)$ has x-intercepts at all points of the form $(\frac{\pi}{12} + k\pi, 0)$, $k \in \mathbb{Z}$ or $(\frac{5\pi}{12} + k\pi, 0)$, $k \in \mathbb{Z}$ $\boxed{1\,\mathsf{A}}$
The set of all x-intercepts of $f(x)$ is

$$\left\{\left(\frac{\pi}{12} + k\pi, 0\right) \middle| k \in \mathbb{Z}\right\} \cup \left\{\left(\frac{5\pi}{12} + k\pi, 0\right) \middle| k \in \mathbb{Z}\right\}$$

$\boxed{\text{Subtotal 6}}$

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Question 3 (11 marks)

Consider the functions $f(x) = \sqrt{x+1}$ and $g(x) = x^2 + 2x - 1$.

(a) What is the maximal domain of $f(x)$? And what is the maximal domain of $g(x)$?

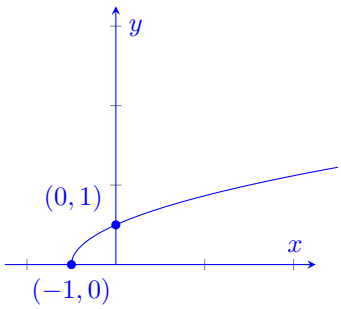
$f(x)$ is defined when $x + 1 \geq 0$. Therefore, the domain of $f(x)$ is $[-1, \infty)$ $\boxed{1\,\text{A}}$

$g(x)$ is a polynomial and, therefore, its domain is $\mathbb{R}$ $\boxed{1\,\text{A}}$

Subtotal 2

(b) Sketch $f(x)$, labelling any important point(s).

$\boxed{1\,\text{A}}$



Subtotal 1

(c) Find the inverse of $f(x)$ and state its domain and range.

The inverse of $f(x)$ has domain $[0, \infty)$ and range $[-1, \infty)$ $\boxed{2\,\text{A}}$

$$y = \sqrt{x+1}$$
$$y^2 = x + 1$$
$$y^2 - 1 = x \quad \boxed{1\,\text{M}}$$

So the inverse of $f(x)$ is $h(y) = y^2 - 1$ $\boxed{1\,\text{A}}$

Subtotal 4

(d) Find an expression for $f(g(x))$.

$f(g(x)) = f(x^2 + 2x - 1) = \sqrt{x^2 + 2x}$ $\boxed{1\,\text{A}}$

Subtotal 1

(e) What is the maximal domain of $f(g(x))$?

$f(g(x))$ is defined when $x^2 + 2x \geq 0$. $\boxed{1\,\text{M}}$.

But the polynomial $x^2 + 2x = x(x+2)$ is an upward-opening parabola with roots $x = 0$ and $x = -2$ $\boxed{1\,\text{J}}$.

Therefore, the domain of $f(g(x))$ is $(-\infty, -2] \cup [0, \infty)$ $\boxed{1\,\text{A}}$

Subtotal 3

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Question 4 (14 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \dfrac{2-x}{5+x^2}$.

(a) Find all x and y intercepts of the function.

- y-intercept at $f(0) = \dfrac{2-0}{5+0^2} = \dfrac{2}{5}$. So y-intercept is at $(0, \frac{2}{5})$ $\boxed{1\,\text{A}}$
- x-intercept at $f(x) = 0$, so
  $0 = \dfrac{2-x}{5+x^2} \implies 2 - x = 0 \implies x = 2.$
  There is an x-intercept at $(2, 0)$ $\boxed{1\,\text{A}}$

Subtotal 2

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative.

Partition domain by roots:

- $f(0) > 0$ so $f(x)$ positive on $(-\infty, 2)$
- $f(3) < 0$ so $f(x)$ negative on $(2, \infty)$ $\boxed{1\,\text{M}}$ $\boxed{1\,\text{A}}$

Subtotal 2

(c) Evaluate and simplify $f'(x)$.

Use quotient rule $f = \frac{u}{v}$: $\boxed{1\,\text{M}}$

$$u = 2 - x \qquad u' = -1$$
$$v = 5 + x^2 \qquad v' = 2x$$

So we have:
$$f'(x) = \frac{u'v - uv'}{v^2} = \frac{-(5+x^2) - (2-x)2x}{(5+x^2)^2}$$
$$= \frac{x^2 - 4x - 5}{(5+x^2)^2} \quad \boxed{1\,\text{A}}$$
$$= \frac{(x-5)(x+1)}{(5+x^2)^2} \quad \boxed{1\,\text{A}} \text{ (correct factorisation)}$$

Subtotal 3

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(d) Find the location of the critical points of $f(x)$.

Critical points occur when $f'(x)$ is not defined or when $f'(x) = 0$. $f'(x)$ is defined for all $x \in \mathbb{R}$ $\boxed{1\,\mathsf{J}}$, so we need only solve $f'(x) = 0$:

$$0 = \frac{(x-5)(x+1)}{(5+x^2)^2} \implies x = 5 \text{ or } x = -1$$

Therefore $f(x)$ has critical points at $x = -1$ and $x = 5$ $\boxed{2\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing.

Partition the domain by critical points and check the sign of the derivative on the different intervals:

- $f'(-2) > 0$ so $f(x)$ is increasing on $(-\infty, -1)$
- $f'(0) < 0$ so $f(x)$ is decreasing on $(-1, 5)$
- $f'(6) > 0$ so $f(x)$ is increasing on $(5, \infty)$ $\boxed{1\,\mathsf{M}}$ $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 2}}$

(f) Find and classify the location of any local extrema of the function.

There is a local maximum at $x = -1$ and a local minimum at $x = 5$ by the first derivative test. $\boxed{2\,\mathsf{A}}$

$\boxed{\text{Subtotal 2}}$

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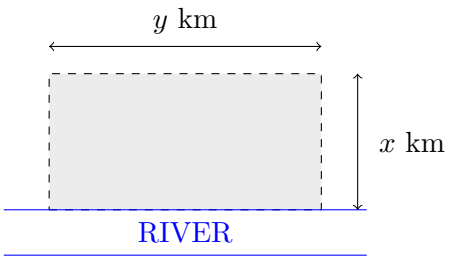
Question 5 (9 marks)

A farmer wants to use fencing wire to fence a rectangular piece of land, using a river as one of the sides (so no fencing is needed on that side), as shown in the picture below:



(a) What is the total length of fencing wire he will need for the job? (Give your answer in terms of x and y)

The farmer will need a total of $(2x + y)$ km of fencing wire. 1 A

Subtotal 1

(b) The farmer wants to use a total of 4km of fencing wire for the job. Use this information to write an expression for y in terms of x.

We are given that $2x + y = 4$. Therefore, $y = 4 - 2x$. 1 A

Subtotal 1

(c) What is the area of the enclosed land in the situation described above? Leave your answer in terms of x only. (*You can assume that the farmer is using the 4km of fencing wire as in part (b)*)

The area of the rectangular piece of land is:

$$A = xy = x(4 - 2x) \text{ km}^2 \quad \boxed{1\,\text{A}}$$

Subtotal 1

(d) What are the dimensions x and y that give the maximum enclosed area? What is the maximum enclosed area?

We want to maximise $A(x) = x(4 - 2x) = 4x - 2x^2$.
We calculate the derivative: $A'(x) = 4 - 4x = 4(1 - x)$ 1 A
Critical points occur either when $A'(x)$ is not defined or when $A'(x) = 0$. In this case, $A'(x)$ is a polynomial and is, therefore, defined for all x. So we only need to solve:

$$A'(x) = 4(1 - x) = 0 \quad \boxed{1\,\text{M}} \implies x = 1$$

Subtotal 2

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There is only one critical point at $x = 1$. $\boxed{1\,\mathsf{A}}$
To check that a maximum occurs at $x = 1$, we check the sign of $A'(x)$ for $x < 1$ and for $x > 1$. $\boxed{1\,\mathsf{M}}$
For $x < 1$, we have $A'(x) = 4(1-x) > 0$, so the function is increasing for $x < 1$.
Similarly, for $x > 1$, we have $A'(x) = 4(1-x) < 0$ and so the function is decreasing for $x > 1$.
Therefore, at $x = 1$ the function changes from increasing to decreasing and so there is a maximum at $x = 1$. The corresponding y-value is $y = 4 - 2x = 4 - 2 = 2$.
So the maximum enclosed area is given when $x = 1$ km and $y = 2$ km. The maximum area is $A = 4(1) - 2(1)^2 = 2$ km $\boxed{2\,\mathsf{A}}$

$\boxed{\text{Subtotal 4}}$

Question 6 (5 marks)

A melting snowball (which can be assumed to always be spherical in shape) is decreasing in volume at a constant rate of 8 cm^3/min. Find the rate at which the radius is changing when the snowball has a radius of 4cm. (*Hint: The volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$*)

We know that $\dfrac{dV}{dt} = -8$ cm^3/min $\boxed{1}$, and we want to find $\dfrac{dr}{dt}$ when $r = 4$ cm. By chain rule: $\boxed{1\,\mathsf{M}}$

$$\frac{dV}{dt} = \frac{dV}{dr}\frac{dr}{dt}$$

We have that $V = \frac{4}{3}\pi r^3$, so $\dfrac{dV}{dr} = 4\pi r^2$ $\boxed{1\,\mathsf{A}}$
Substituting into the previous expression gives:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \implies \frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$$

Finally, substituting the values $r = 4$ and $\dfrac{dV}{dt} = -8$ gives: $\boxed{1\,\mathsf{M}}$

$$\frac{dr}{dt} = \frac{1}{4\pi(4)^2}(-8) = -\frac{1}{8\pi}\text{cm/min}$$

The radius is changing at a rate of $-\frac{1}{8\pi}$cm/min when the snowball has a radius of 4cm. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 5}}$

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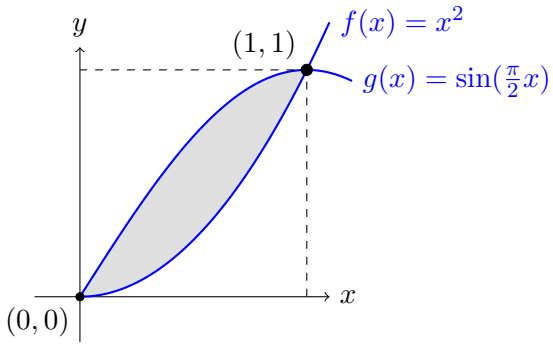
Question 7 (7 marks)

Consider the functions $f(x) = x^2$ and $g(x) = \sin\left(\frac{\pi}{2}x\right)$, with graphs as shown below:



(a) Show that the two functions intersect at $x = 0$ and at $x = 1$.

We need only verify that $f(0) = g(0)$ and $f(1) = g(1)$:

$$f(0) = 0^2 = 0, \qquad f(1) = 1^2 = 1$$

$$g(0) = \sin(0) = 0 \qquad g(1) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$\implies f(0) = g(0) \qquad \implies f(1) = g(1)$$

$\boxed{1\,\text{M}}$ $\boxed{1\,\text{A}}$ $\boxed{\text{Subtotal 2}}$

(b) Find the value of the shaded area.

The shaded area is given by:

$$\begin{aligned}
A &= \int_0^1 \left(\sin\left(\frac{\pi}{2}x\right) - x^2\right)\, dx \quad \boxed{1\,\text{M}} \\
&= \left[-\frac{2}{\pi}\cos\left(\frac{\pi}{2}x\right) - \frac{1}{3}x^3\right]_0^1 \quad \boxed{2\,\text{A}} \\
&= \left[-\frac{2}{\pi}\cos\left(\frac{\pi}{2}\right) - \frac{1}{3}\right] - \left[-\frac{2}{\pi}\cos(0) - \frac{1}{3}\cdot 0\right] \\
&= \left[-0 - \frac{1}{3}\right] - \left[-\frac{2}{\pi} - 0\right] \quad \boxed{1} \text{ (correct cosine values)} \\
&= \frac{2}{\pi} - \frac{1}{3} \quad \boxed{1\,\text{A}}
\end{aligned}$$

$\boxed{\text{Subtotal 5}}$

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Question 8 (8 marks)

For this question, a decimal approximation is not required. You may leave your final answer in simplified fractional form.

The conditional probability formula says that, for two events $A, B \in \Omega$ with $P(A) \neq 0$,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

For any MAST10012 student, the probability that they forget the formula during the exam is $p = \frac{2}{3}$. We randomly choose 4 MAST10012 students and define X to be the number of those students that forget the formula during the exam.

(a) Calculate the probability that exactly 2 of those 4 students forget the conditional probability formula during the exam.

Here $X \sim B(4, \frac{2}{3})$ $\boxed{1\,\mathsf{M}}$. We want to calculate $P(X = 2)$:

$$\begin{aligned} P(X = 2) &= \binom{4}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^2 \quad \boxed{1} \\ &= \frac{4!}{2!2!}\left(\frac{4}{9}\right)\left(\frac{1}{9}\right) \\ &= \frac{4 \cdot 3 \cdot 2!}{2!2!}\left(\frac{4}{81}\right) \\ &= 6\left(\frac{4}{81}\right) = \frac{8}{27} \end{aligned}$$

The probability that 2 out of those 4 students forget the conditional probability formula during the exam is $\frac{8}{27}$ $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

(b) What is the probability that at least one of the students forgets the formula during the exam?

$$\begin{aligned} P(X \geq 1) &= 1 - P(X < 1) \quad \boxed{1\,\mathsf{M}} \\ &= 1 - P(X = 0) \\ &= 1 - \binom{4}{0}\left(\frac{2}{3}\right)^0\left(\frac{1}{3}\right)^4 \\ &= 1 - \frac{4!}{4!0!}\left(\frac{1}{3}\right)^4 \\ &= 1 - \left(\frac{1}{81}\right) = \frac{80}{81} \end{aligned}$$

The probability that at least one of the students forgets the formula during the exam is $\frac{80}{81}$ $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 2}}$

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(c) What is the probability that exactly two students forget the formula during the exam, given that at least one student forgets it?

Here, we need to use conditional probability. 1 M. We want to calculate $P(X = 2|X \geq 1)$:

$$\begin{aligned} P(X = 2|X \geq 1) &= \frac{P((X = 2) \cap (X \geq 1))}{P(X \geq 1)} \\ &= \frac{P(X = 2)}{P(X \geq 1)} \quad \boxed{1} \\ &= \frac{\frac{8}{27}}{\frac{80}{81}} \\ &= \frac{8 \cdot 81}{27 \cdot 80} \\ &= \frac{3}{10} \end{aligned}$$

Subtotal 3

The probability that exactly two students forget the formula during the exam, given that at least one student forgets it is $\frac{3}{10}$. 1 A

Question 9 (5 marks)

In how many ways can the letters of the word MATHS be arranged

(a) if all the letters are used?

If the 5 different letters are to be used, we have permutations of 5 elements, so the total number of ways we can arrange the letters is $5! = 120$ 1 A

Subtotal 1

(b) if we form 3-letter words?

We first need to select which three letters we use, so we have $\binom{5}{3} = \frac{5!}{3!2!} = 10$ ways to do this. 1 M

For each of these 3-letter groups, we have $3! = 6$ possible orderings of the letters. 1 M

These two choices are independent, therefore multiplication principle applies. 1 M

We have a total of $6 \times 10 = 60$ ways of forming 3-letter words. 1 A

Subtotal 4

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Question 10 (5 marks)

The probability density function for a random variable X is

$$f(x) = \begin{cases} 1 & a < x < 2 \\ 3 - x & 2 \leq x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that we must have $a = \frac{3}{2}$.

We know that for $f(x)$ to be a probabillity density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx = 1$. Therefore: $\boxed{1\,\text{M}}$

$$\begin{aligned} 1 = \int_{-\infty}^{\infty} f(x)\,dx &= \int_{-\infty}^{a} 0\,dx + \int_{a}^{2} 1\,dx + \int_{2}^{3} (3-x)\,dx + \int_{3}^{\infty} 0\,dx \\ &= \Big[x\Big]_a^2 + \Big[3x - \frac{x^2}{2}\Big]_2^3 \ \boxed{1\,\text{A}} \\ &= \Big[2 - a\Big] + \Big[9 - \frac{9}{2}\Big] - \Big[6 - \frac{4}{2}\Big] \\ &= \frac{5}{2} - a \ \boxed{1\,\text{A}} \end{aligned}$$

So we must have $1 = \frac{5}{2} - a \implies a = \frac{5}{2} - 1 = \frac{3}{2}$

$\boxed{\text{Subtotal 3}}$

(b) Find $P(X < 2)$.

$$\begin{aligned} P(X < 2) = \int_{-\infty}^{2} f(x)\,dx &= \int_{-\infty}^{\frac{3}{2}} 0\,dx + \int_{\frac{3}{2}}^{2} 1\,dx \ \boxed{1\,\text{M}} \\ &= \Big[x\Big]_{\frac{3}{2}}^{2} \\ &= 2 - \frac{3}{2} \\ &= \frac{1}{2} \ \boxed{1\,\text{A}} \end{aligned}$$

$\boxed{\text{Subtotal 2}}$

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Question 11 (8 marks)

The amount of water Mr. Green uses to water his garden is normally distributed with mean 100 litres and standard deviation 10 litres.

(a) On 20% of occasions, it takes him less than k litres of water to water the garden. What is the value of k?

Let X be the amount of water Mr. Green uses to water his garden. We know that the mean is 100, with standard deviation 10, so $X \sim N(100, 10)$. We want to find k such that $P(X < k) = 0.2$.
We start by considering a standard normal $Z \sim N(0, 1)$. We want to find z such that $P(Z < z) = 0.2$. $\boxed{1}$
Using symmetry, we have $P(Z < z) = 1 - P(Z < -z)$. Hence we seek $P(Z < -z) = 0.8$. $\boxed{1}$
From the table, this gives $-z = 0.84$, so $z = -0.84$.
Converting to our non-standard normal gives:

$$\begin{aligned} -0.84 &= \frac{k - 100}{10} \quad \boxed{1\,\mathsf{M}} \\ -8.4 &= k - 100 \\ 100 - 8.4 &= k \end{aligned}$$

Hence the value of k is 91.6 $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 4}}$

(b) He wants to be environmentally friendly and aims to use between 90 and 95 litres to water his garden. What percentage of times is he hitting that goal?

Convert to a standard normal first.We want:

$$\begin{aligned} P(90 < X < 95) &= P\left(\frac{90 - 100}{10} < Z < \frac{95 - 100}{10}\right) \quad \boxed{1\,\mathsf{M}} \\ &= P(-1 < Z < -0.5) \\ &= P(0.5 < Z < 1) \text{ (by symmetry)} \quad \boxed{1\,\mathsf{M}} \\ &= P(Z < 1) - P(Z < 0.5) \quad \boxed{1\,\mathsf{M}} \\ &= 0.8413 - 0.6915 \text{ (from the table)} \\ &= 0.1498 \end{aligned}$$

Therefore, he is meeting the goal in about 14.98% of the occasions. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 4}}$

End of Exam — Total Available Marks = 106

Turn the page for appended material

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Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

- **Functions:**
  - Lines

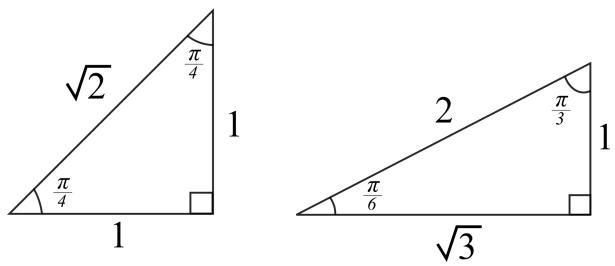
$$y = mx + c \qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**



- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$

$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$

$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$

$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$

$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$

$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$

$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$

$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$

$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$

$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$

$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

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- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a} [f(x) + g(x)] = K + L \qquad \lim_{x\to a} [f(x) - g(x)] = K - L$$

$$\lim_{x\to a} [f(x) \times g(x)] = K \times L \qquad \lim_{x\to a} \left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$

$$\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a} [f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$

$$\lim_{x\to a} cf(x) = cK$$

$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$

$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{a\to x} \frac{f(a) - f(x)}{a - x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

- **Binomial:** For $X \sim B(n,p)$:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$\mathbb{E}(X) = np$$

$$\mathrm{Var}(X) = np(1-p)$$

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The standard normal distribution table for $P(X < x)$

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000